

Scanner Service Implementation Guide

Version 2.1 • 52 service families and IP forwarding
Implemented policy, evidence, report flow and acceptance limits

Updated from the attached v2.0 service document after Laptop and central code changes.

Includes the complete service list, ports, credential/network boundaries and exact process diagrams.

Source build 1.5.3-v45.7. Native feed-backed checks and live acceptance are explicitly distinguished.

SOURCE REVIEW + SCANNER RELEASE 1.5.3-v45.7

06 October 2026 | Separate execution paths, shared evidence rules

Contents and diagram key

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Diagram key

 Process	 Gate / branch	 Retained evidence
 Key / crypto	 Gap / failure	 External workflow

Solid arrows show implemented transitions. Dashed arrows show an explicitly external or conditional path. A gap is evidence of a limitation; it does not establish that the requested data never existed.

Implementation / Scope and evidence contract

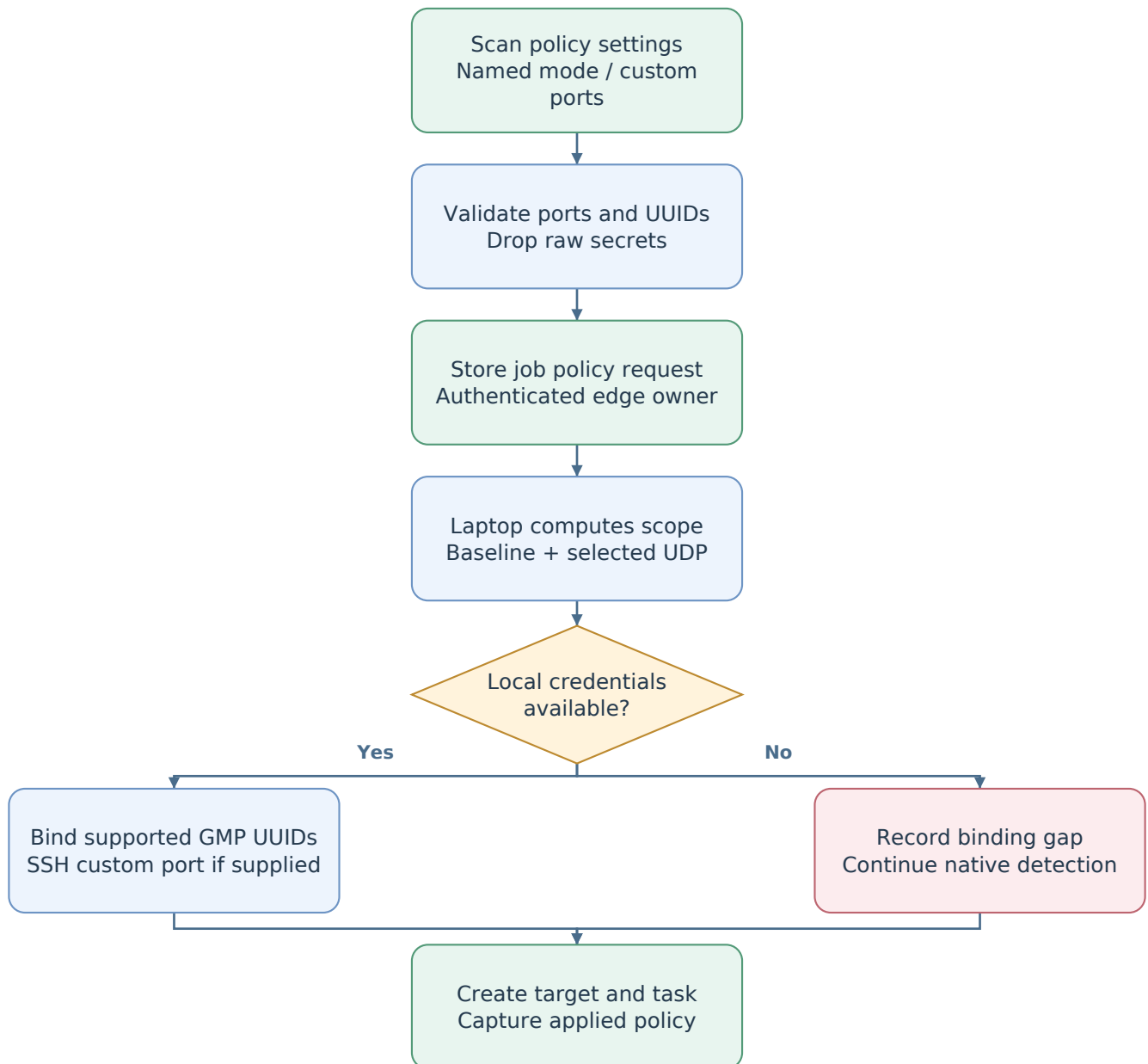
Updated from the supplied 18-page v2.0 service guide. This revision describes actual code changes in source build 1.5.3-v45.7 and preserves visible implementation limits.

Requirement	Implemented behavior / acceptance boundary
52 services + configuration	All 52 ordered service/product entries are registered as SVC-01...SVC-52. IP forwarding is CFG-IP-FORWARDING, an additional configuration family.
Scope and custom ports	Full TCP is default. Named catalog modes include 88 TCP ports; priority UDP includes 30 ports. Custom TCP/UDP ports add scope and can be configured per policy or locally.
Checks	Native Greenbone/OpenVAS feed performs protocol/product checks. This release adds routing/audit/evidence handling, not 52 independent custom vulnerability engines.
Credentials	Explicit local GMP UUIDs bind SSH, SMB, SNMP and ESXi credentials; optional SSH credential port is supported. Bound is distinct from successful authentication.
Evidence states	Family findings/observations require source references or native context. Missing evidence is not-tested; unsupported network collection is explicit. No pass is inferred from a quiet port.
Reporting	UI coverage detail; per-host 52-family CSV plus forwarding; JSON task policy audit; PDF/DOCX coverage appendix, including merged cases.
Live validation	Windows/Docker/Greenbone coverage/performance, credentialed product checks and raw Nessus parity still require real authorized target fixtures.

Implementation: Source guide v2.0; scanner_service_policy.py; service_catalog.json; scanner_agent_jobs.py

Service policy / Request to actual target scope

A policy is snapshotted when the central job is created, so later policy edits cannot change the claimed job's request.

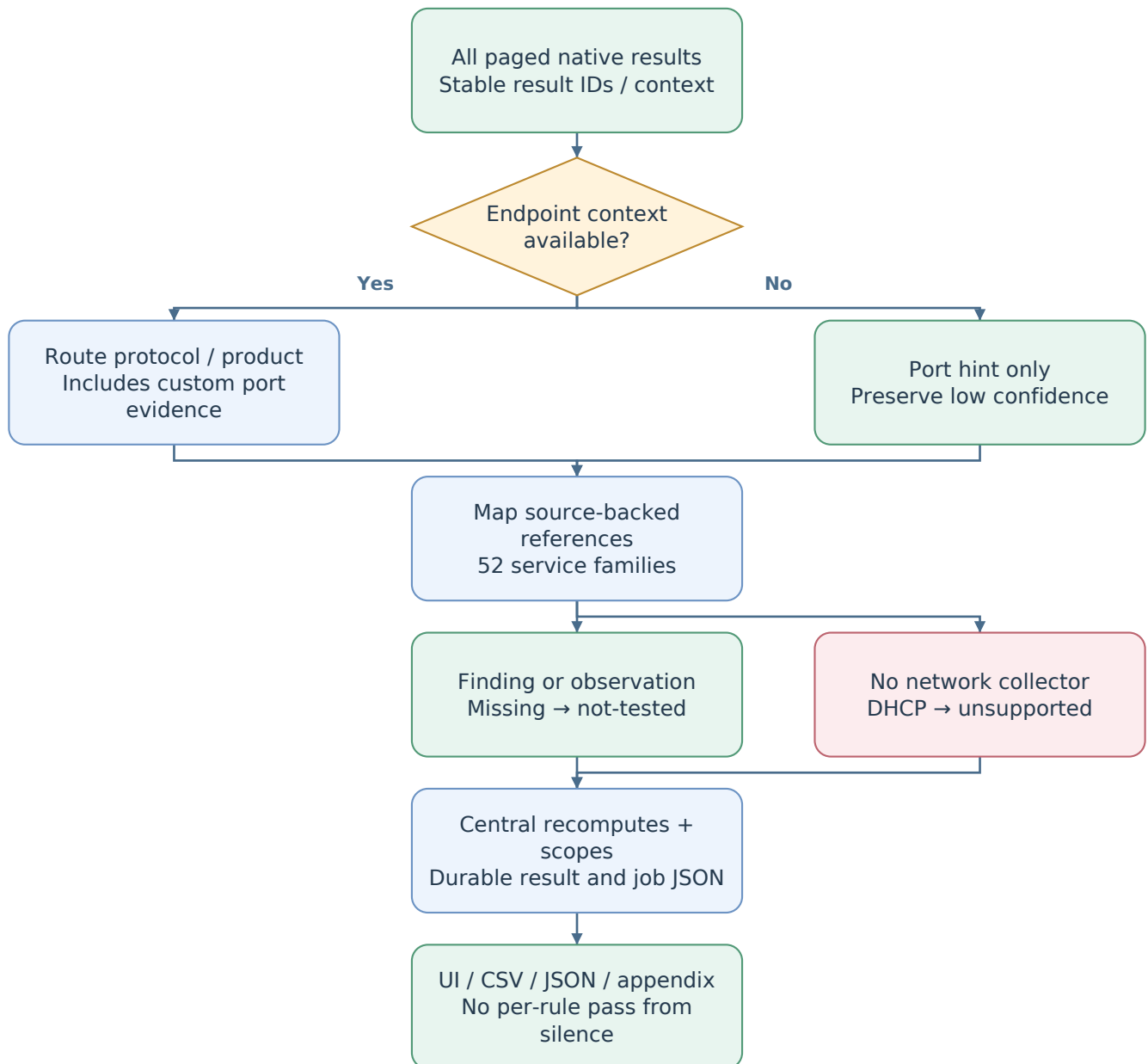


Credential boundary Credentials must already exist in the local Greenbone vault. A central credential_ref is not a local GMP UUID. WMI/WinRM and product-specific authenticated inventory depend on supported native feed methods; no new universal collector is implied.

Implementation: backend/app/routers/vuln.py; scanner_agent_jobs.py; agent/service_policy.py; gmp_local.py

Service evidence / Native report to family status

A service label is a source-backed observation. Port numbers alone do not identify a product or establish vulnerability.

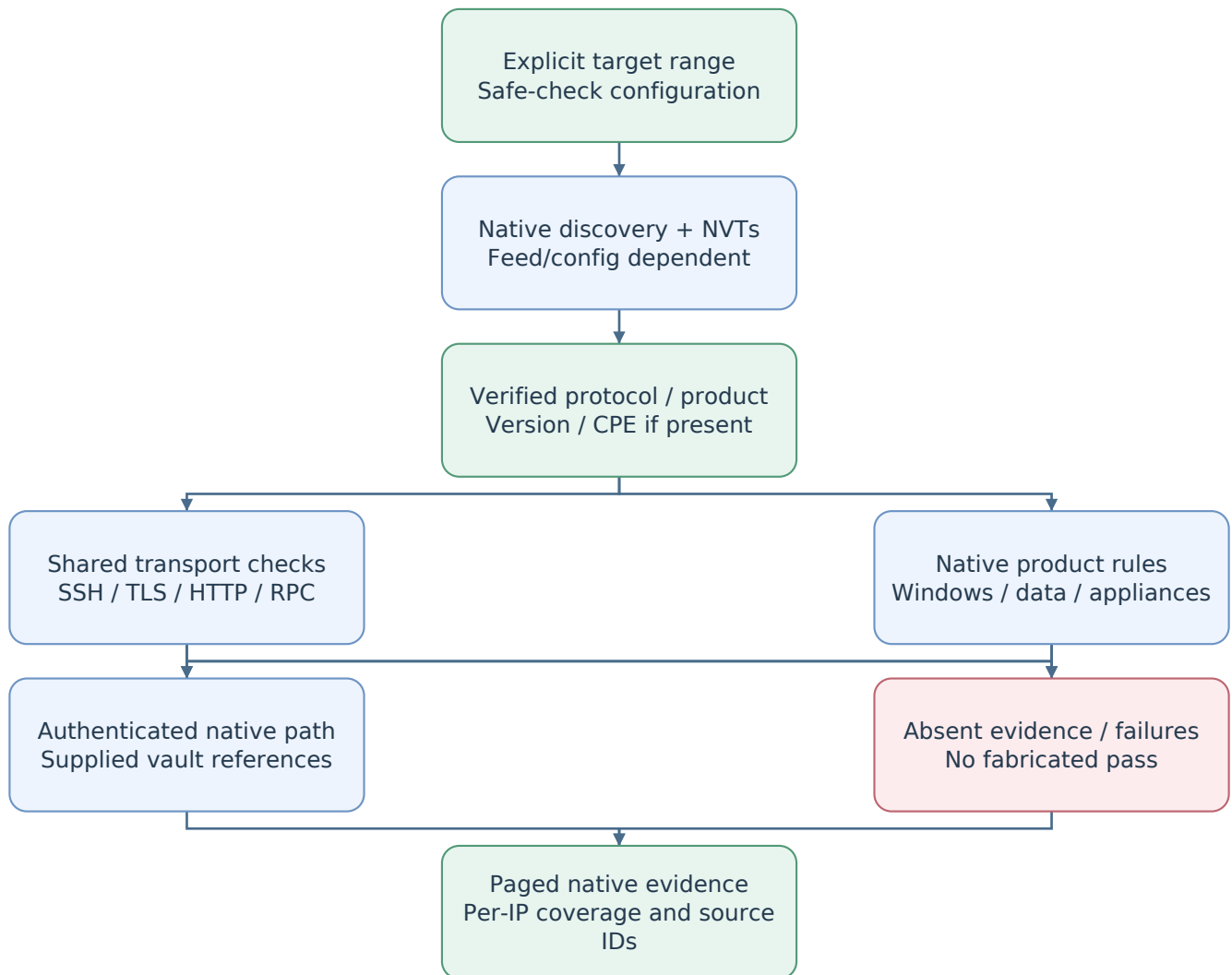


Evidence limit Native plugin-name references can associate a family without proving a precise fingerprint; they remain references, not a fabricated version/CPE. Missing score is explicitly marked. Configuration IP forwarding requires source-backed role/configuration evidence.

Implementation: agent/service_policy.py; gmp_local.py; backend/app/services/scanner_agent_jobs.py; vuln_report_export.py

Execution / Detection and native check families

Registered scope reaches the same native Full and fast configuration. Endpoint context routes evidence to shared protocols and product families.



Shared checks TLS/STARTTLS and SSH can support many products. The native feed controls check execution and deduplication. Per-SNI TLS cache or a new dedicated legacy-protocol engine was not added; target compatibility must be validated.

Device scope Vendor, NAS or mail-vault support requires actual discovery and available native rules. A vendor name in the registry is not proof of validated product coverage.

Implementation: gmp_local.py; service_policy.py; native Greenbone feed and configuration

Policy / Inputs, defaults and immutable audit

The central launch stores sanitized policy settings in `orchestration_json`. The claimed job receives this request; each created native task records what was actually applied.

Setting / field	Meaning and limits
port_profile / mode	full (default), fast, common, vuln, nessus or nexus. All named non-Full modes retain the guide baseline and versioned catalog.
udp_profile	priority (default), full or off. Off removes UDP for every mode; Full requests 1–65535 UDP. Priority is additive baseline + transport extensions + existing catalog.
custom_tcp_ports / custom_udp_ports	Comma-separated ports/ranges or lists. Validate 1–65535, reject malformed/reversed/Boolean values. More than 4096 custom ports requires Full mode.
Local defaults	SCAN_TCP_CUSTOM_PORTS and SCAN_UDP_CUSTOM_PORTS. Legacy UDP_PORT_RANGE adds ports and cannot erase the baseline.
gmp_credential_refs	Only ssh/smb/snmp/esxi UUIDs. Local vault lookup/binding failure is recorded; unauthenticated detection continues. Raw passwords/URLs/tokens are dropped from transport.
ssh_credential_port	One validated port, default 22. Set it when an authorized SSH account is bound on a nonstandard port.
Task snapshot	Task/target/config/scanner IDs, effective/excluded port range and counts, guide version, feed versions, applied preferences, credential status, start time and SHA-256.
Resume	Persist snapshots immediately in progress; restore only policies whose entire target list belongs to this job. Old tasks without recorded policies stay unknown, not reconstructed from current defaults.

Implementation: `service_policy.py`; `gmp_local.py`; `main.py`; `backend/app/routers/vuln.py`

Service catalog / 01-06

Native feed-dependent scope. Hint ports assist discovery; an open port alone does not identify this service or prove that its requested checks passed.

ID / service and meaning	Hint ports / transport	Intended native check scope
SVC-01 SMB / CIFS Server Message Block / Common Internet File System	TCP: 139,445 UDP: —	Remote: SMBv1; signing; shares and permissions Authenticated: supplied-user share permissions
SVC-02 RPC Remote Procedure Call	TCP: 135 UDP: — Discovered RPC ports	Remote: Endpoint mapping; dynamic RPC services; build checks Authenticated: authenticated Windows inventory
SVC-03 WMI Windows Management Instrumentation	TCP: 135 UDP: — Discovered RPC ports	Remote: Route Windows RPC evidence Authenticated: patch/configuration inventory
SVC-04 WinRM Windows Remote Management	TCP: 5985,5986 UDP: —	Remote: Authentication and endpoint TLS Authenticated: Windows configuration/inventory
SVC-05 RDP Remote Desktop Protocol	TCP: 3389 UDP: 3389	Remote: Security negotiation; TLS; NLA; version Authenticated: host patch/configuration
SVC-06 SSH: OpenSSH / Dropbear Secure Shell server products	TCP: 22 UDP: —	Remote: Product/algorithms; exact vendor advisory checks Authenticated: package/backport and SSH configuration

Implementation status Registered and routed in code. Actual native rule results require feed/configuration, protocol/product identity, access and report evidence. Credentialed/network coverage remains not-tested unless proven.

Implementation: agent/service_catalog.json; service_policy.py; Aetheris-Service-Coverage-List.csv

Service catalog / 07-12

Native feed-dependent scope. Hint ports assist discovery; an open port alone does not identify this service or prove that its requested checks passed.

ID / service and meaning	Hint ports / transport	Intended native check scope
SVC-07 FTP / FTPS File Transfer Protocol / FTP secured with TLS	TCP: 20,21,990 UDP: —	Remote: Banner; bounded anonymous access; explicit/implicit TLS Authenticated: server configuration/access policy
SVC-08 SFTP SSH File Transfer Protocol	TCP: 22 UDP: —	Remote: Reuse SSH transport evidence Authenticated: SFTP permissions/configuration
SVC-09 HTTP Hypertext Transfer Protocol	TCP: 80,8000,8080 UDP: — Custom endpoints supported	Remote: Web server/app identity; headers/exposure; route product checks Authenticated: authenticated application checks
SVC-10 HTTPS / TLS Hypertext Transfer Protocol Secure / Transport Layer Security	TCP: 443,8443 UDP: — Custom endpoints supported	Remote: Certificates; protocol/cipher acceptance; STARTTLS Authenticated: configuration context where supplied
SVC-11 SMTP / SMTPS Simple Mail Transfer Protocol / SMTP secured with TLS	TCP: 25,465,587 UDP: —	Remote: Banner; authentication; relay policy; STARTTLS Authenticated: mail configuration/access policy
SVC-12 POP3 / POP3S Post Office Protocol v3 / POP3 secured with TLS	TCP: 110,995 UDP: —	Remote: Product; cleartext authentication; STARTTLS Authenticated: mail configuration/access policy

Implementation status Registered and routed in code. Actual native rule results require feed/configuration, protocol/product identity, access and report evidence. Credentialed/network coverage remains not-tested unless proven.

Implementation: agent/service_catalog.json; service_policy.py; Aetheris-Service-Coverage-List.csv

Service catalog / 13-18

Native feed-dependent scope. Hint ports assist discovery; an open port alone does not identify this service or prove that its requested checks passed.

ID / service and meaning	Hint ports / transport	Intended native check scope
SVC-13 IMAP / IMAPS Internet Message Access Protocol / IMAP secured with TLS	TCP: 143,993 UDP: —	Remote: Product; authentication; STARTTLS Authenticated: mail configuration/access policy
SVC-14 DNS Domain Name System	TCP: 53 UDP: 53	Remote: Protocol; recursion/transfer policy; server identity Authenticated: server configuration
SVC-15 SNMP Simple Network Management Protocol	TCP: Configured / none fixed UDP: 161	Remote: Version; configured community/access; firmware Authenticated: approved device/community inventory
SVC-16 Telnet Remote terminal protocol	TCP: 23 UDP: —	Remote: Protocol/product; cleartext remote access Authenticated: device access configuration
SVC-17 TFTP Trivial File Transfer Protocol	TCP: Configured / none fixed UDP: 69	Remote: Protocol; approved-file read policy Authenticated: approved filename/access context
SVC-18 LDAP Lightweight Directory Access Protocol	TCP: 389 UDP: 389	Remote: Bind policy; anonymous access; STARTTLS Authenticated: directory configuration

Implementation status Registered and routed in code. Actual native rule results require feed/configuration, protocol/product identity, access and report evidence. Credentialed/network coverage remains not-tested unless proven.

Implementation: agent/service_catalog.json; service_policy.py; Aetheris-Service-Coverage-List.csv

Service catalog / 19-24

Native feed-dependent scope. Hint ports assist discovery; an open port alone does not identify this service or prove that its requested checks passed.

ID / service and meaning	Hint ports / transport	Intended native check scope
SVC-19 LDAPS LDAP secured with TLS	TCP: 636 UDP: —	Remote: TLS and directory bind/access Authenticated: directory configuration
SVC-20 Kerberos Network authentication protocol	TCP: 88 UDP: 88	Remote: Protocol/realm identity Authenticated: credentialed AD policy
SVC-21 AD Global Catalog Active Directory Global Catalog	TCP: 3268,3269 UDP: —	Remote: Directory exposure/bind; secure endpoint TLS Authenticated: directory configuration
SVC-22 Microsoft SQL Server Structured Query Language database	TCP: 1433 UDP: —	Remote: TDS; build/version; encryption/access Authenticated: patch/configuration/permissions
SVC-23 MySQL / MariaDB Database server products	TCP: 3306 UDP: —	Remote: Handshake; authentication/access; TLS; advisories Authenticated: packages/settings/permissions
SVC-24 PostgreSQL Database server product	TCP: 5432 UDP: —	Remote: Protocol; SSL negotiation; authentication/patch context Authenticated: packages/settings/permissions

Implementation status Registered and routed in code. Actual native rule results require feed/configuration, protocol/product identity, access and report evidence. Credentialed/network coverage remains not-tested unless proven.

Implementation: agent/service_catalog.json; service_policy.py; Aetheris-Service-Coverage-List.csv

Service catalog / 25-30

Native feed-dependent scope. Hint ports assist discovery; an open port alone does not identify this service or prove that its requested checks passed.

ID / service and meaning	Hint ports / transport	Intended native check scope
SVC-25 Oracle Database Oracle relational database	TCP: 1521 UDP: —	Remote: Listener/service identification; access/encryption Authenticated: database/listener patch inventory
SVC-26 MongoDB Document database	TCP: 27017 UDP: —	Remote: Authentication/authorization; TLS; version/exposure Authenticated: server configuration/roles
SVC-27 Redis Remote Dictionary Server	TCP: 6379 UDP: —	Remote: Authentication/ACL; TLS; exposure/version Authenticated: read-only config/inventory
SVC-28 Elasticsearch Search and analytics product	TCP: 9200,9300 UDP: —	Remote: HTTP/transport identity; authentication; TLS Authenticated: security/build/patch context
SVC-29 VNC Virtual Network Computing	TCP: 5900 UDP: — Custom endpoints supported	Remote: RFB; authentication/encryption; product advisories Authenticated: configured endpoint policy
SVC-30 VMware ESXi / vCenter Hypervisor and management products	TCP: 443,902 UDP: 902	Remote: Management/API identity; access; TLS; advisories Authenticated: vendor patch/inventory

Implementation status Registered and routed in code. Actual native rule results require feed/configuration, protocol/product identity, access and report evidence. Credentialed/network coverage remains not-tested unless proven.

Implementation: agent/service_catalog.json; service_policy.py; Aetheris-Service-Coverage-List.csv

Service catalog / 31-36

Native feed-dependent scope. Hint ports assist discovery; an open port alone does not identify this service or prove that its requested checks passed.

ID / service and meaning	Hint ports / transport	Intended native check scope
SVC-31 Docker API Docker Application Programming Interface	TCP: 2375,2376 UDP: —	Remote: API exposure; auth/client certificates; version Authenticated: daemon configuration
SVC-32 Kubernetes API Kubernetes Application Programming Interface	TCP: 6443 UDP: —	Remote: API identity; anonymous access; TLS Authenticated: RBAC and authenticated context
SVC-33 kubelet Kubernetes node agent	TCP: 10250 UDP: —	Remote: Node API exposure; authorization; TLS Authenticated: node configuration
SVC-34 Jenkins Automation server	TCP: 8080 UDP: — Custom endpoints supported	Remote: Core/plugin identity; admin exposure; advisories Authenticated: plugin/configuration inventory
SVC-35 GitLab / Git Development platform / version control	TCP: 22,80,443,9418 UDP: — Custom endpoints supported	Remote: GitLab identity/access/advisories; Git exposure; reuse SSH/TLS Authenticated: platform configuration
SVC-36 RabbitMQ Message broker	TCP: 5672,15672 UDP: — Custom endpoints supported	Remote: AMQP/management; access; TLS; version Authenticated: permissions/broker settings

Implementation status Registered and routed in code. Actual native rule results require feed/configuration, protocol/product identity, access and report evidence. Credentialed/network coverage remains not-tested unless proven.

Implementation: agent/service_catalog.json; service_policy.py; Aetheris-Service-Coverage-List.csv

Service catalog / 37-42

Native feed-dependent scope. Hint ports assist discovery; an open port alone does not identify this service or prove that its requested checks passed.

ID / service and meaning	Hint ports / transport	Intended native check scope
SVC-37 Apache Kafka Event streaming product	TCP: 9092 UDP: —	Remote: Broker identity; auth/ACLs; listener TLS Authenticated: broker configuration
SVC-38 Memcached Memory caching service	TCP: 11211 UDP: 11211	Remote: TCP/UDP identity; exposure; version Authenticated: transport/access policy
SVC-39 Apache Tomcat Java application server	TCP: 8080,8443 UDP: — Custom endpoints supported	Remote: Product/advisories; manager exposure; TLS Authenticated: server/component patch context
SVC-40 JBoss / WildFly Java application servers	TCP: 8080,9990 UDP: — Custom endpoints supported	Remote: Application/admin identity; access; advisories Authenticated: server/component patch context
SVC-41 IIS Internet Information Services	TCP: 80,443 UDP: — Custom endpoints supported	Remote: HTTP/TLS identity and exposure Authenticated: Windows features/patches
SVC-42 Apache HTTPD Apache HTTP Server daemon	TCP: 80,443 UDP: — Custom endpoints supported	Remote: Version/modules; configuration/exposure; vendor patches Authenticated: package/backport context

Implementation status Registered and routed in code. Actual native rule results require feed/configuration, protocol/product identity, access and report evidence. Credentialed/network coverage remains not-tested unless proven.

Implementation: agent/service_catalog.json; service_policy.py; Aetheris-Service-Coverage-List.csv

Service catalog / 43-48

Native feed-dependent scope. Hint ports assist discovery; an open port alone does not identify this service or prove that its requested checks passed.

ID / service and meaning	Hint ports / transport	Intended native check scope
SVC-43 nginx Web server product	TCP: 80,443 UDP: — Custom endpoints supported	Remote: Version/build; configuration/exposure; vendor patches Authenticated: package/backport context
SVC-44 Oracle WebLogic Oracle WebLogic Server	TCP: 7001,7002 UDP: — Custom endpoints supported	Remote: Version/patch; console exposure; auth/TLS Authenticated: server patch/configuration
SVC-45 IBM WebSphere IBM WebSphere Application Server	TCP: Configured / none fixed UDP: — Custom endpoints supported	Remote: Configured app/admin identity; version; auth/TLS Authenticated: server patch/configuration
SVC-46 IPsec / IKE Internet Protocol Security / Internet Key Exchange	TCP: Configured / none fixed UDP: 500,4500	Remote: Negotiation; device identity; vendor/configuration Authenticated: device configuration
SVC-47 NTP Network Time Protocol	TCP: Configured / none fixed UDP: 123	Remote: Protocol/version; permitted query policy Authenticated: device configuration
SVC-48 DHCP Dynamic Host Configuration Protocol	TCP: Configured / none fixed UDP: 67,68	Remote: Network-scoped server identity Authenticated: network broadcast/relay collector

Implementation status Registered and routed in code. Actual native rule results require feed/configuration, protocol/product identity, access and report evidence. Credentialed/network coverage remains not-tested unless proven.

Implementation: agent/service_catalog.json; service_policy.py; Aetheris-Service-Coverage-List.csv

Service catalog / 49-52

Native feed-dependent scope. Hint ports assist discovery; an open port alone does not identify this service or prove that its requested checks passed.

ID / service and meaning	Hint ports / transport	Intended native check scope
SVC-49 IPP Internet Printing Protocol	TCP: 631 UDP: —	Remote: Printer identity/access; supported TLS Authenticated: device/firmware inventory
SVC-50 JetDirect HP network printing technology	TCP: 9100 UDP: —	Remote: Device identity/exposure; no arbitrary raw print payload Authenticated: device/firmware inventory
SVC-51 NFS Network File System	TCP: 2049 UDP: 2049	Remote: RPC discovery; export/access policy; version Authenticated: approved export permission context
SVC-52 rpcbind Remote Procedure Call binding service	TCP: 111 UDP: 111 Discovered RPC ports	Remote: RPC program enumeration; auxiliary endpoints Authenticated: host/RPC configuration

Implementation status Registered and routed in code. Actual native rule results require feed/configuration, protocol/product identity, access and report evidence. Credentialed/network coverage remains not-tested unless proven.

Implementation: agent/service_catalog.json; service_policy.py; Aetheris-Service-Coverage-List.csv

UDP / Baseline, extensions and network collection

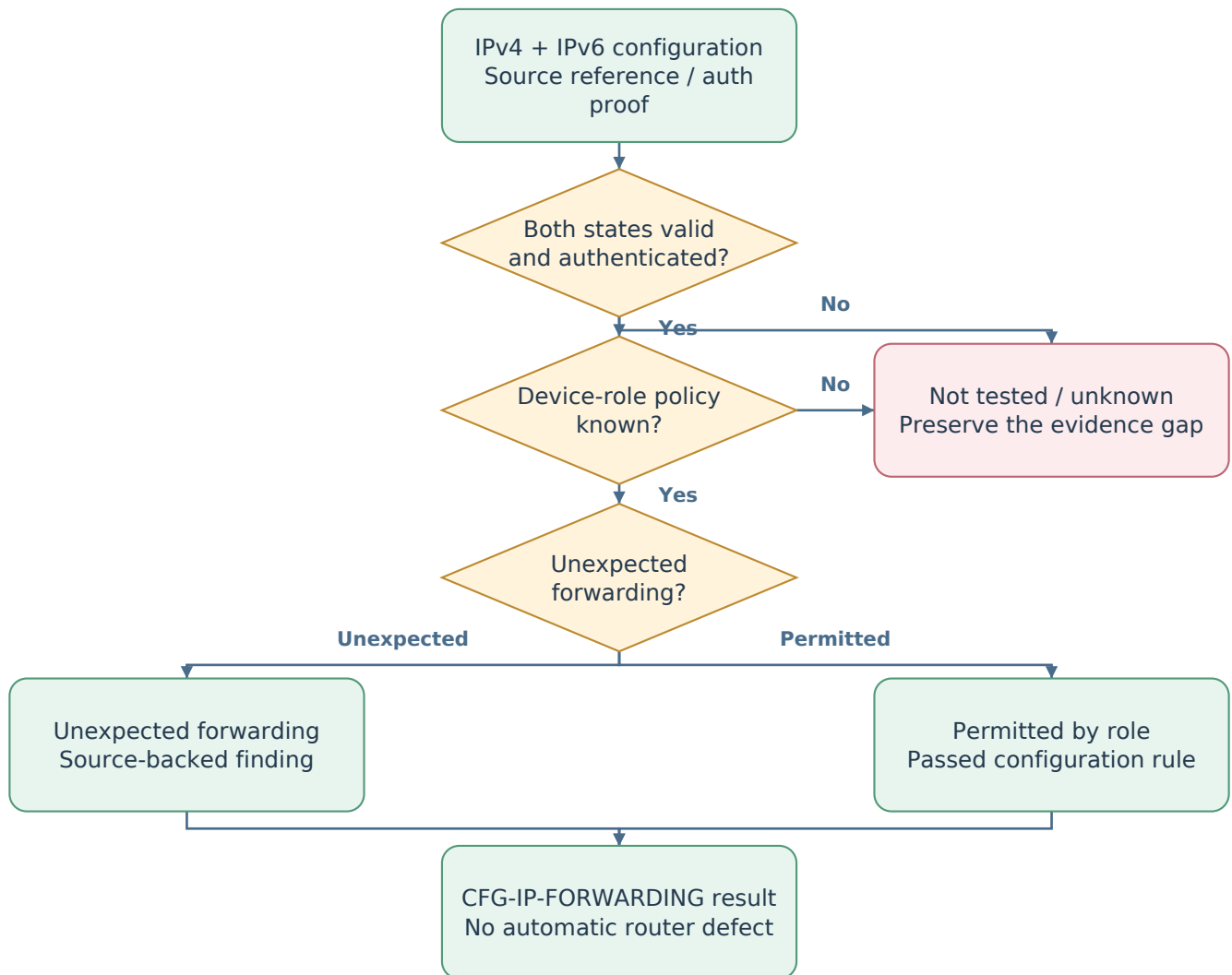
Numeric ports are not service counts. Priority scanning is explicit scope; UDP timeout/filtering is not a passed vulnerability test.

Group	Ports / current behavior
Guide baseline (17)	53, 67, 68, 69, 123, 137, 161, 162, 500, 514, 520, 623, 1434, 1900, 4500, 5353, 11211
Guide transport extensions (6)	88 Kerberos; 111 rpcbind; 389 CLDAP; 902 VMware; 2049 NFS; 3389 RDP.
Existing additional scope (7)	138 NetBIOS; 1701 L2TP; 1812/1813 RADIUS; 4789 VXLAN; 5004 RTP; 5060 SIP. Union gives 30 default priority UDP ports.
Full / Off	Full UDP requests all 65,535 ports. Off disables UDP without silently changing TCP. Custom additions respect the selected mode.
Network collector gap	DHCP discovery needs broadcast/relay/network scope. SSDP/mDNS may also need network-scoped discovery. No separate DHCP/SSDP/mDNS collector was added to the per-IP agent.
RPC / custom listeners	Full TCP covers dynamic/custom TCP numbers; Fast/catalog mode needs supplied additions for ports outside its named list. Native RPC enumeration depends on reachable services and feed support.
Printing boundary	JetDirect 9100 remains a listed endpoint. No custom raw printing payload was added. Native printer-check behavior must be validated on authorized fixtures.

Implementation: service_catalog.json; effective_policy; gmp_local_effective_scan_policy

Configuration / IP forwarding evaluation

This code evaluates supplied authenticated configuration; it does not acquire IPv4/IPv6 state by a new SSH/WMI/device collector.



Collector limit The ordinary per-IP service export reports forwarding not-tested without configuration evidence. The evaluator is available for a supported source-backed collector; no title-only or credential-free configuration claim is made.

Implementation: `service_policy.evaluate_ip_forwarding`; `service_catalog.json`

Reference findings / Eight guide families

These are labels and comparator cases from the supplied guide, not independently verified vulnerability declarations. Raw scanner/plugin evidence is required for exact parity.

Guide reference	Evidence needed before a finding / parity claim
SSLv2 / SSLv3	Protocol negotiation on the actual endpoint/SNI or STARTTLS context; record unsupported legacy runtime separately. Native feed evidence must identify the condition.
OpenSSH < 7.2 X11	Actual product version/build, enabled condition, vendor backport/advisory applicability, native plugin/CVE and source output.
Dropbear < 2016.72	Product/version plus applicable vendor advisory/build evidence; do not classify from TCP 22 alone.
SMB unprivileged shares	Share and permission evidence for the tested supplied user/access context, not merely an open SMB port.
SWEET32	Supported 64-bit block cipher negotiation and relevant endpoint context. Retain cipher and protocol evidence.
OpenSSH prior to 10.3 label	Treat the supplied export title as a comparison label. Obtain raw plugin/advisory/version applicability; do not invent a universal fixed version or severity.
Dropbear < 2013.59	Exact native plugin, product/build/version, vendor advisory/CVE and detection proof.
IP forwarding	Authenticated IPv4/IPv6 state and device-role policy, or verified native network evidence. Legitimate routing is not automatically a defect.

Missing original material The source guide notes that raw Nessus export/plugin IDs, CVEs, ports/vectors and policy details were absent. Exact accuracy/severity parity remains unverified.

Implementation: Supplied guide v2.0; native finding payload and scanner_service_policy.py

Evidence / States, confidence and score provenance

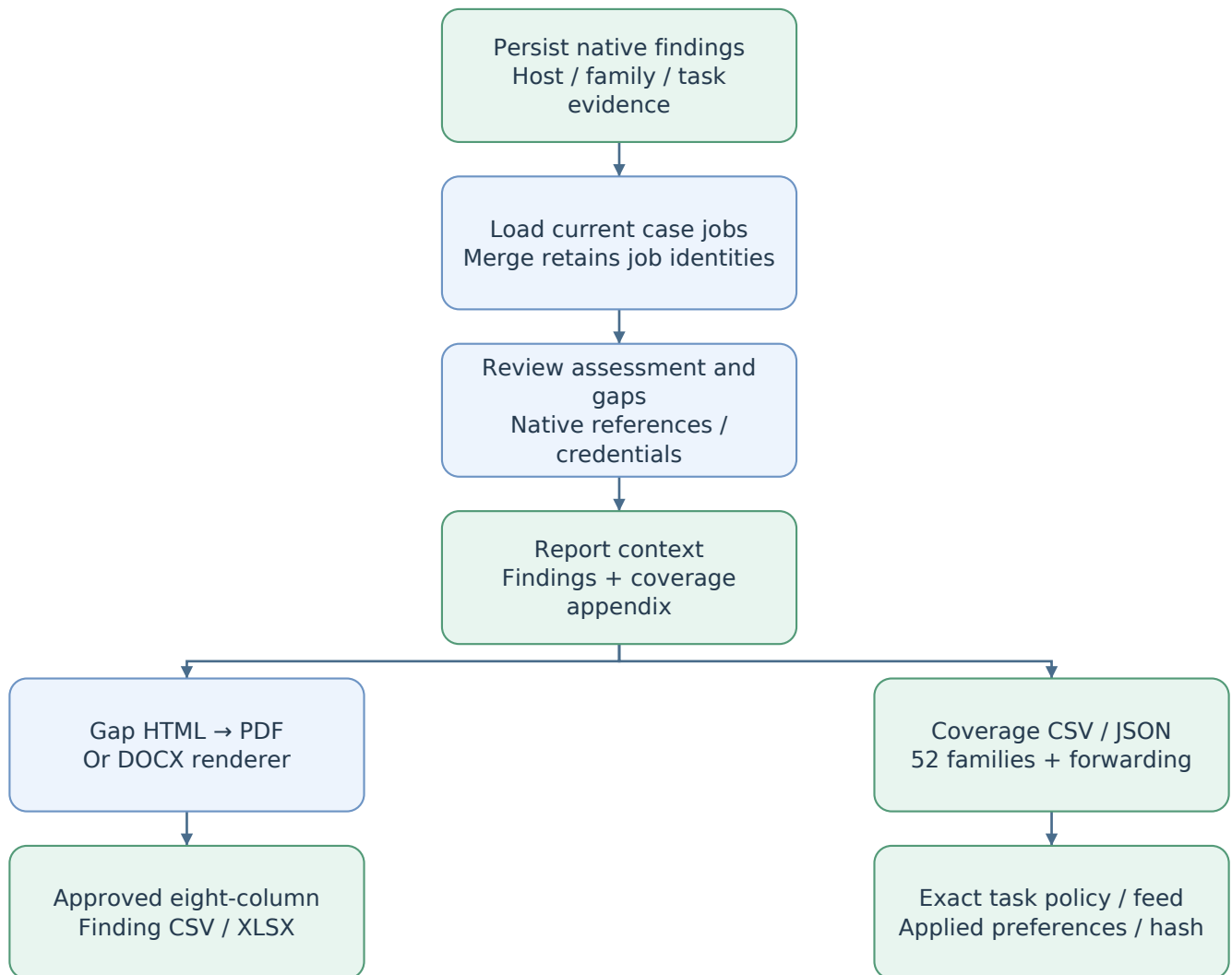
A native Done status is engine completion, not a universal pass for every service or authenticated configuration.

Evidence item	Implemented treatment
Finding / observation	Positive native score with a matching source-backed family reference → finding; native context/reference without a scored condition → observation.
Missing / unsupported	No matching native evidence → not-tested. DHCP without network collection → unsupported. The family rollup does not imply independent per-check success.
Endpoint confidence	Port-only hints remain port_hint_only. Actual explicit service/product/CPE yields native_engine_context. No product/version is guessed from a default port.
Source identity	Result IDs, NVT OIDs/families, host, transport, port, supplied product/version/CPE and raw description remain linked. Central recomputes and scopes service rollups.
Scores	Finite numeric CVSS v2/v3/v4 candidates and raw native severity retained. Missing numeric score is marked score_available=false; it is distinct from a measured zero.
Severity	Central configured CVSS basis determines technical severity; default v3.1 bands use Critical ≥9, High ≥7, Medium ≥4, Low >0. Finding titles do not impose floors.
Contextual priority	Published custom weighted score is separate from native Tenable VPR. Native VPR/EPSS are preserved when supplied; missing threat data is not fabricated from CVSS.
Execution vs rule coverage	Host warnings/degraded port scans cannot become clean from partial=false. Family coverage_complete stays false unless actual rule success is established; a quiet report is not a 52-family clean certificate.

Implementation: gmp_local.py; scanner_agent_jobs.py; aetheris_severity.py; vuln_risk.py

Reporting / Evidence to client output

The same persisted job evidence feeds Scan Details and the client appendix. Finding-only solution exports retain their approved layout.



Interpretation Appendix summaries show findings/observations/not-tested/unsupported counts. CSV lists every family per host; JSON retains source references and policy snapshots. Reports remain exportable for incomplete scans with visible limitations.

Implementation: `vuln_report_export.py`; `gap_assessment_report.py`; `gap_report_template.py`; `ReportsExportPage.tsx`

Performance / Throughput and bounded recovery

The guide's 15–18 IP/hour figure is a benchmark target, not a guaranteed three-minute scan. Preserve service scope and evidence while tuning hardware admission.

Control	Code behavior / measurement
Global IP capacity	Five-worker operational floor with adaptive ceiling (default up to six). Critical CPU/RAM/thermal pressure pauses new admissions; already running native tasks finish.
Per-IP engine budget	Separate one-host tasks. Honor configured checks (default eight, capped at sixteen), native VT timeout 320 s and scanner-plugin budget 36,000 s.
Discovery	Parallel connect probes are observational and do not narrow mandatory scope. No answering ports does not delete SSH/TLS or other target services.
Report harvest	Metadata first, paged native results; stable source-ID deduplication only. Parser/count/report failures remain visible instead of discarding low/informational evidence.
Progress	Complete assigned-IP snapshots with restored monotonic event sequence. Job row lock and owner fencing prevent overwrites or delayed previous-task regressions.
Capacity estimate	Throughput $\approx 60 \times$ concurrent IPs / mean scan minutes. Five-six workers need roughly 16.7–24 mean minutes per IP for 15–18 IP/hour, assuming sustained workload and no overhead bottleneck.
Measure on target	Record feed/config/hash, per-IP runtime, CPU/RAM/thermal pressure, native row counts, timeout/plugin errors and credential gaps. Avoid lowering scope or forcing 100% to hit the rate.

Implementation: agent/capacity.py; ip_semaphore.py; main.py; gmp_local.py; target_progress.py

Deployment / Updated source and commands

Extract the supplied updated archives into their existing project/scanner folders while preserving local environment, runtime identity and acquired evidence.

Where	Action / result
Central source	Use updated polaron-disk-mobile-serial.zip. From the project root run powershell -ExecutionPolicy Bypass -File .\scripts\Apply-Scanner-Integration.ps1
Laptop source	Use Aetheris-Laptop-Scanner-v1.5.3.zip. From laptop-scanner run powershell -ExecutionPolicy Bypass -File .\Apply-Scanner-v1.5.3.ps1
Preserved data	Update rebuilds/recreates application or agent containers. It does not delete Greenbone/feed/database volumes, local .env, tokens or existing LAN fingerprint.
Policy example	settings_json: port_profile="fast", udp_profile="priority", custom_tcp_ports=[2222,9443], gmp_credential_refs={ssh:"local-GMP-UUID"}, ssh_credential_port=2222.
Readiness fix	Launch query now selects openvas_ready and agent_status_detail. A ready edge scanner launches; a false/unknown readiness is blocked with its specific detail.
Coverage exports	Scan Details shows service status and task policy. Reports page adds Service coverage CSV. API: /api/cases/{case_id}/service-coverage (JSON) or .csv.
Old source cleanup	Delivered packages exclude backup trees, caches, logs, compiled/dist dependencies, real scanner secrets and runtime identities. Current release files are authoritative.

Implementation: Apply-Scanner-v1.5.3.ps1; scripts/Apply-Scanner-Integration.ps1; release manifests

Validation / Completed and target-host acceptance

Source tests and document inspection are complete release checks. Live acquisition and scanner behavior require the real authorized hardware/network.

Area	Status / acceptance work
Automated regressions	Guide scope/custom ports, UDP modes, invalid scope, secret filtering, credential references, immutable task audit, evidence routing, forwarding role, progress sequence, readiness launch and export/appendix fixtures. Final counts are in the release manifest.
Central integration	PostgreSQL-compatible WASM tests exercise migrations, heartbeat isolation, target ownership/exclusions and progress/severity SQL. TypeScript/Vite Vuln and Unified builds validate the UI.
Syntax / packages	Python 3.11 AST and changed PowerShell grammar checks; canonical agent JSON/Python parity; ZIP integrity and secret/cache exclusion; PDF bounds/bookmarks/text and visual review.
Windows / Greenbone	Run native PowerShell/Docker deploy and real feed-backed scans; confirm feed readiness, source version, all configured ports, dynamic/custom services and native errors.
All 52 families	Use authorized positive/negative/product-version/permission fixtures and supplied credentials. Match plugin IDs/CVEs and source evidence, not just port-open status.
Raw Nessus parity	Obtain the original scanner export and policy settings; compare exact host/port/product/plugin/vector/severity evidence before claiming parity.
Forensic live acceptance	Separate Disk/Mobile flow PDFs describe current source paths. Live phone/GPU verification remains pending; encrypted WhatsApp backups still require a matching validated key.
Known external gaps	No new DHCP/SSDP/mDNS network collector, dedicated IPv4/IPv6 configuration acquisition, per-SNI TLS cache/legacy engine, or independently validated 52-rule suite is claimed.

Implementation: SCANNER-INTEGRATION-V153-RELEASE.md; SCANNER-INTEGRATION-V153-MANIFEST.json

References / Source and protocol documentation

The supplied guide defines required service scope. Official protocol/metric references support the implementation boundaries; they do not certify a live target scan.

Source	Reference / trace
Supplied service guide	Aetheris_Scanner_Service_Implementation_Guide(1).pdf, v2.0, 18 pages; 52 ordered service rows plus IP forwarding. Updated here as v2.1 implementation status.
Machine-readable registry	agent/service_catalog.json and matching backend/app/services/service_catalog.json; agent/service_policy.py and scanner_service_policy.py.
GMP target API	https://greenbone.github.io/python-gvm/api/gmpv227.html — create_target credential UUID and SSH-port parameters; get_credential and feed methods.
Greenbone GMP	https://docs.greenbone.net/API/GMP/gmp-22.7.html
Nmap version detection	https://nmap.org/book/vscan.html
CVSS v3.1	https://www.first.org/cvss/v3-1/specification-document
Tenable severity / metrics	https://docs.tenable.com/nessus/Content/ConfigureIndividualScanSeverity.htm https://docs.tenable.com/nessus/Content/RiskMetrics.htm
Separate flow references	Disk-Process-Flows.pdf; Mobile-Process-Flows.pdf; Vulnerability-Scanner-Process-Flows.pdf; companion editable scanner .odt.

Implementation: docs/process-flows / reproducible JSON and vector diagram generator